

SmartPack: Current Project State & Audit

Comprehensive Status Assessment, Working Features, ML Metrics, and Technical Roadmap
SIH Problem Statement 26236: Smart Food-Packaging Recommendation System

Overall System Status:	FUNCTIONAL & LOCALLY VERIFIED (PROTOTYPE READY)
Backend API:	FastAPI Service (All 9 Endpoints Active & Passing Automated Tests)
Frontend Web Application:	React 18 + TypeScript + Tailwind CSS (Fully Built with Real-Time Sliders)
Machine Learning Model:	Trained Scikit-Learn Random Forest Ensemble (91.83% Accuracy / 0.9187 R ²)
Integrated Databases:	14 CSV Datasets (496 ICMR Foods, 102 FSSAI Schedule IV Mappings)
Deployment Status:	Configured for Cloud Hosting (Render Blueprint `render.yaml` + Vite Static)

1. Subsystem Implementation Status

Every architectural tier of the SmartPack system has been evaluated and confirmed operational:

Subsystem	Component	Current State	Key Features & Verification
Backend API	FastAPI Engine	100% Complete	Endpoints: /api/analyze, /api/health, /api/foods, /api/ml/status, /api/explain/gemini. Automated test suite passing.
ML Core	Random Forest Model	100% Trained	Dual-head regressor & classifier serialized at models/packaging_suitability_model.pkl. 7,200 dataset rows, 18 tabular features.
Rule Engine	Requirement & Constraints	100% Complete	Computes oxygen, moisture, mechanical & sealability demands. Enforces hard safety filters (FSSAI Reg 4(4) and IS 5837).
Scoring System	Hybrid Scoring Engine	100% Complete	Linear 6-factor combination (ML 30%, Barrier 20%, Compatibility 15%, Shelf-life 15%, Mechanical 10%, Sustainability 10%).
Frontend UI	React 18 Dashboard	100% Complete	Features ICMR autocomplete, interactive sliders, requirement gauges, Top-3 cards, comparison matrix, and developer drawer.
AI Narrative	Gemini 1.5 Flash LLM	Integrated	Dynamic biochemical rationale generator citing FSSAI statutory regulations for every recommended packaging option.
Data Layer	Physical Barrier DB	Operational (Gaps Identified)	FSSAI Schedule IV & ICMR foods fully linked. Polymer-level OTR/WVTR data gaps identified in source dataset (see Section 4).

2. Machine Learning Model Audit

The serialized model in models/packaging_suitability_model.pkl was evaluated on held-out test data (25% split, 1,800 rows):

Evaluation Parameter	Audited Metric	Engineering Benchmark & Interpretation
Model Architecture	RandomForestRegressor & Classifier	Ensemble of 100 decision trees (max depth 8) balancing bias and variance.
Continuous R ² Score	0.9187 (91.87%)	High variance explained across multi-attribute requirement interactions.
Mean Absolute Error (MAE)	0.0348 (~3.5 points on 100 scale)	Very tight calibration to domain suitability scoring rules.
Root Mean Squared Error	0.0448	Low penalty for large outliers, ensuring consistent ranking.
Classification Accuracy	91.83%	Successfully discriminates between suitable and unsuitable options.
Precision / Recall / F1	P: 85.86% R: 84.66% F1: 85.26%	Balanced performance with minimal false positive recommendations.

3. Operational Capabilities in the Current Build

The following end-to-end workflows are active, tested, and operational:

- **Dynamic ICMR Food Search:** Real-time querying of 496 foods from ICMR IFCT 2017, auto-populating baseline moisture, fat %, and food group.
- **Multi-Factor Vulnerability Modeling:** Calculates physical and biological deterioration risks (lipid rancidity, deliquescence, mechanical transit shock, MAP gas needs).
- **Hard Safety Regulatory Interlock:** Automatically rejects prohibited packaging (e.g. acidic food metal corrosion under IS 5837, direct contact recycled plastics).
- **Configurable Multi-Factor Scoring:** Transparently blends ML predictions with barrier, compatibility, shelf-life, and circularity weights.
- **Side-by-Side Comparison Matrix:** Renders full engineering specifications for Top-3 choices.
- **Developer Telemetry & Funnel Logging:** Slide-out drawer tracking candidate reduction from 102 candidates down to 3 final selections.

4. Known Bottlenecks, Data Gaps & Transparency Disclosures

In strict compliance with **Section 12 Transparency Rules** and scientific auditing standards, the following data realities and limitations are documented:

Identified Gap / Limitation	Root Cause in Primary Data	Current Behavior & Mitigation Strategy
Barrier Rate Gaps (OTR/WVTR)	Source dataset 08_barrier_properties_dataset.csv only had 2 rows for cellulose film.	System strictly refuses to fabricate numbers. Commodity plastics without test rows display 'Data unavailable'.
Rigid Container Barrier Display	Glass bottles & tin cans are not flexible films, so they lacked film transmission entries in raw CSVs.	Physical reality: Glass & tin are impermeable (OTR/WVTR = 0.0). Recommended to display as '0.0 (Impermeable)'.
Closure Type Defaulting	Table defaulted to 'Heat-Seal / Pouch Crimp' for non-bottle items.	UI will be updated to display exact closures (Lug Cap for glass, Double Seam for tinplate cans).
Prototype Target (Section 12)	No public dataset exists containing 10,000 lab shelf-life test records for all food-package pairs.	ML model trains on rule-derived prototype targets, explicitly labeled 'DERIVED SCORE', not empirical lab proof.

5. Deployment Readiness & Immediate Roadmap

Cloud Deployment Status:

- **Render Blueprint ('render.yaml'):** Fully configured for single-command cloud deployment with automated Python virtual environment setup and Vite frontend build.
- **Automated Test Suite:** Unit and pipeline tests in backend/test_pipeline.py and backend/test_api.py pass with zero errors.
- **Environment Configuration:** .env.example prepared with Google Gemini API key slots and CORS settings.

Immediate Next Engineering Steps:

1. **Standard Barrier Catalog Enhancement:** Map well-established polymer ASTM ranges (PET, PP, HDPE, LDPE, Aluminium foil) and label glass/tin as '0.0 (Impermeable)' to replace 'Not available'.
2. **Dynamic Closure Mapping:** Connect closure types directly to container morphology (crown cork, lug cap, hermetic can seam, pouch heat-seal).
3. **Multi-Lingual Packaging Reports:** Generate exportable FSSAI compliance certificates in regional Indian languages for local food entrepreneurs.

Executive Summary: SmartPack is currently in an advanced, operational prototype stage (v1.0.0-prototype). All core algorithms, ML models, safety interlocks, and frontend interfaces are completely functional, stable, and ready for demonstration, with clear scientific boundaries and audit trails in place.